QPCODE **Reg No: ……**

Name: …….

MAHATMA GANDHI UNIVERSITY ,KOTTAYAM

MGU-UGP(HONOURS)

FIFTH SEMESTER EXAMINATION

(2024 YEAR ADMISSION ONWARDS)

MG5DSECFS310- M-COMMERCE SECURITY

Duration:2hrs **Maximum Marks:70**

PART A

Short Answer Type Questions
Answer any 10 questions
Each question carries 2 marks

1. Describe M-Commerce and mention any two of its features. [Understand][CO1]

2. Explain any two types of M-Commerce transactions. [Understand][CO1]

3. Describe NFC (Near Field Communication) and its role in mobile payments. [Understand][CO1]

4. Explain the difference between authentication and authorization. [Understand][CO2]

5. Discuss Multi-Factor Authentication (MFA) and its importance in M-Commerce. [Understand][CO2]

6. Explain Role-Based Access Control (RBAC) with a suitable example. [Understand][CO2]

7. Identify the components of the CIA Triad and describe their importance in M-Commerce security. [Understand][CO3]

8. Explain the difference between symmetric cryptography and public-key cryptography. [Understand][CO3]

9. Describe Elliptic Curve Cryptography (ECC) and explain its use in M-Commerce. [Understand][CO3]

10. Compare phishing and social engineering attacks in M-Commerce. [Analyse][CO4]

11. Distinguish between secure and insecure communication channels in M-Commerce. [Analyse][CO4]

12. Examine the security risks caused by injection attacks in M-Commerce applications.
[Analyse][CO4]

(10x2=20)

PART B

Short Essay Type Questions
Answer any 6 questions
Each question carries 5 marks

13. Explain the structure of M-Commerce and describe its major components.

[Understand][CO1]

14. Describe the different mobile payment technologies used in M-Commerce, including NFC, QR-code payments and biometric authentication. [Understand][CO1]

15. Describe the different authentication methods used in M-Commerce, including password-based authentication, Multi-Factor Authentication (MFA) and biometric authentication.

[Understand][CO2]

16. Differentiate between RBAC, ABAC and PBAC with suitable examples.

[Understand][CO2]

17. Explain the architecture and different models of mobile payment systems.

[Understand][CO3]

18. Describe the security requirements of mobile payment systems, including confidentiality, integrity, availability, authorization and non-repudiation. [Understand][CO3]

19. Distinguish between phishing, malware, Man-in-the-Middle (MITM) attacks, social engineering and weak authentication based on their impact on M-Commerce transactions.

[Analyse][CO4]

20. Examine the major vulnerabilities in M-Commerce applications, including insecure data storage, injection attacks, insecure input validation, insecure communication channels and Insecure Direct Object References (IDOR). [Analyse][CO4]

(6x5=30)

PART C

Essay Type Questions
Answer any 2 questions
Each question carries 10 marks

21. Explain the concept and structure of M-Commerce. Classify the different types of M-Commerce transactions and discuss the major security challenges associated with M-Commerce. [Understand][CO1]

22. Discuss the importance of authentication and authorization in M-Commerce. Describe password-based authentication, Multi-Factor Authentication (MFA), biometric authentication, RBAC, ABAC and PBAC. [Understand][CO2]

23. Describe the architecture of mobile payment systems and discuss their major security requirements, including confidentiality, integrity, availability, authorization and non-repudiation. Explain the role of symmetric cryptography, public-key cryptography and Elliptic Curve Cryptography (ECC) in securing mobile payments. [Understand][CO3]

(2x10=20)